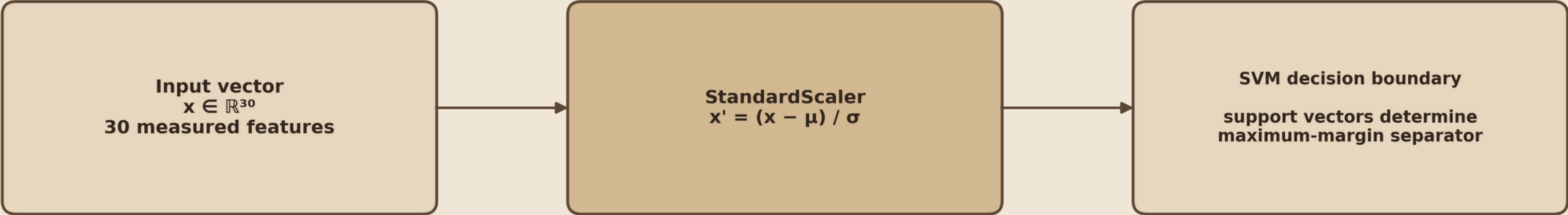
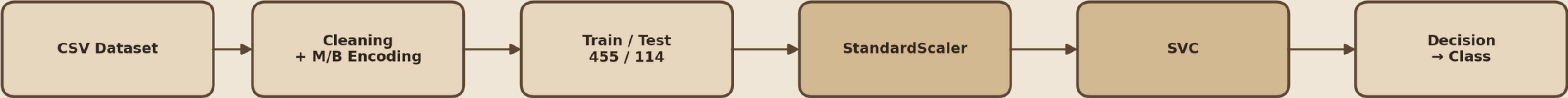


SVM

Classification Report:				
	precision	recall	f1-score	support
B	0.97	1.00	0.99	71
M	1.00	0.95	0.98	43
accuracy			0.98	114
macro avg	0.99	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

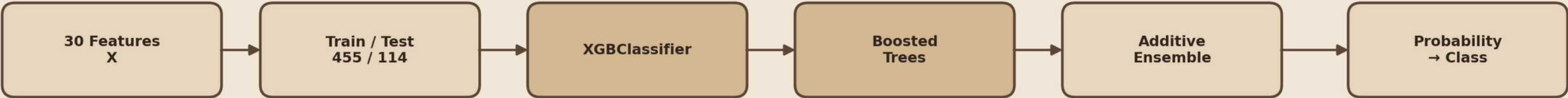
Detailed SVM Architecture



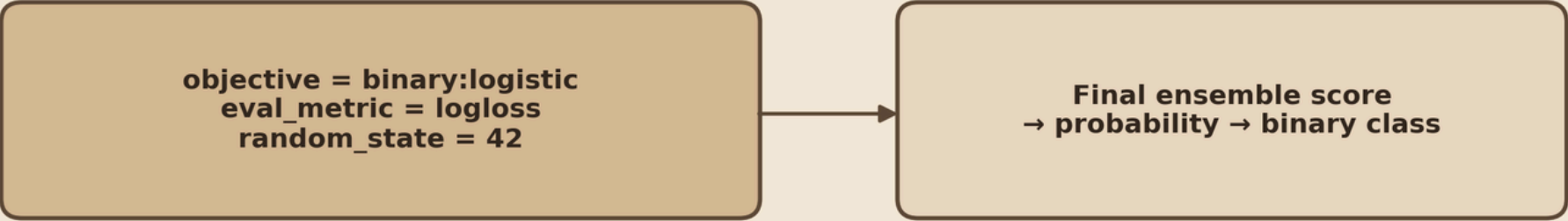
XGBoost

XGBoost Classification Report:					
		precision	recall	f1-score	support
	B	0.96	0.99	0.97	71
	M	0.98	0.93	0.95	43
	accuracy			0.96	114
	macro avg	0.97	0.96	0.96	114
	weighted avg	0.97	0.96	0.96	114

Detailed XGBoost Architecture



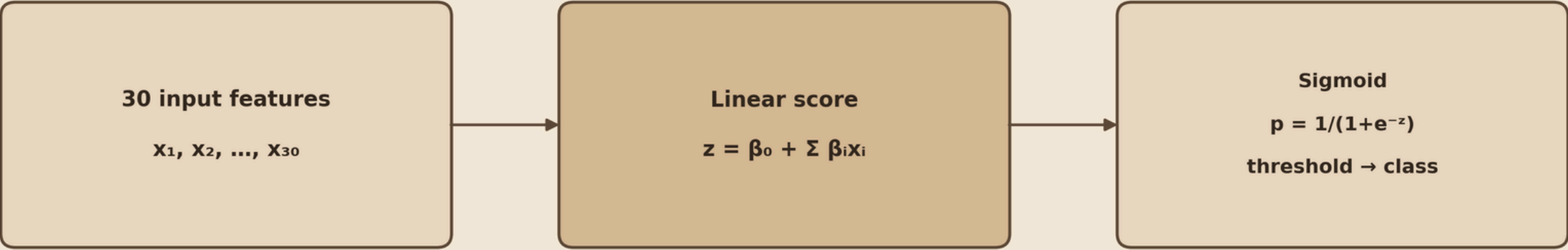
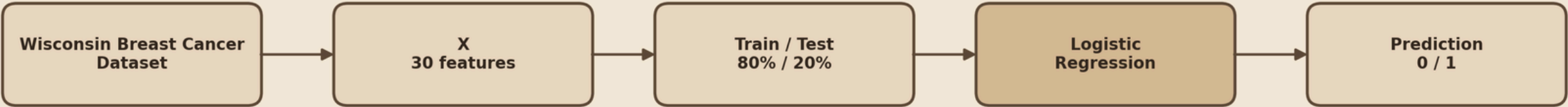
Sequential boosting: later trees correct errors left by earlier trees



Logistic Regression

Classification Report:					
	precision	recall	f1-score	support	
	0	0.91	0.91	0.91	45
	1	0.94	0.94	0.94	69
	accuracy			0.93	114
	macro avg	0.93	0.93	0.93	114
	weighted avg	0.93	0.93	0.93	114

Detailed Logistic Regression Architecture

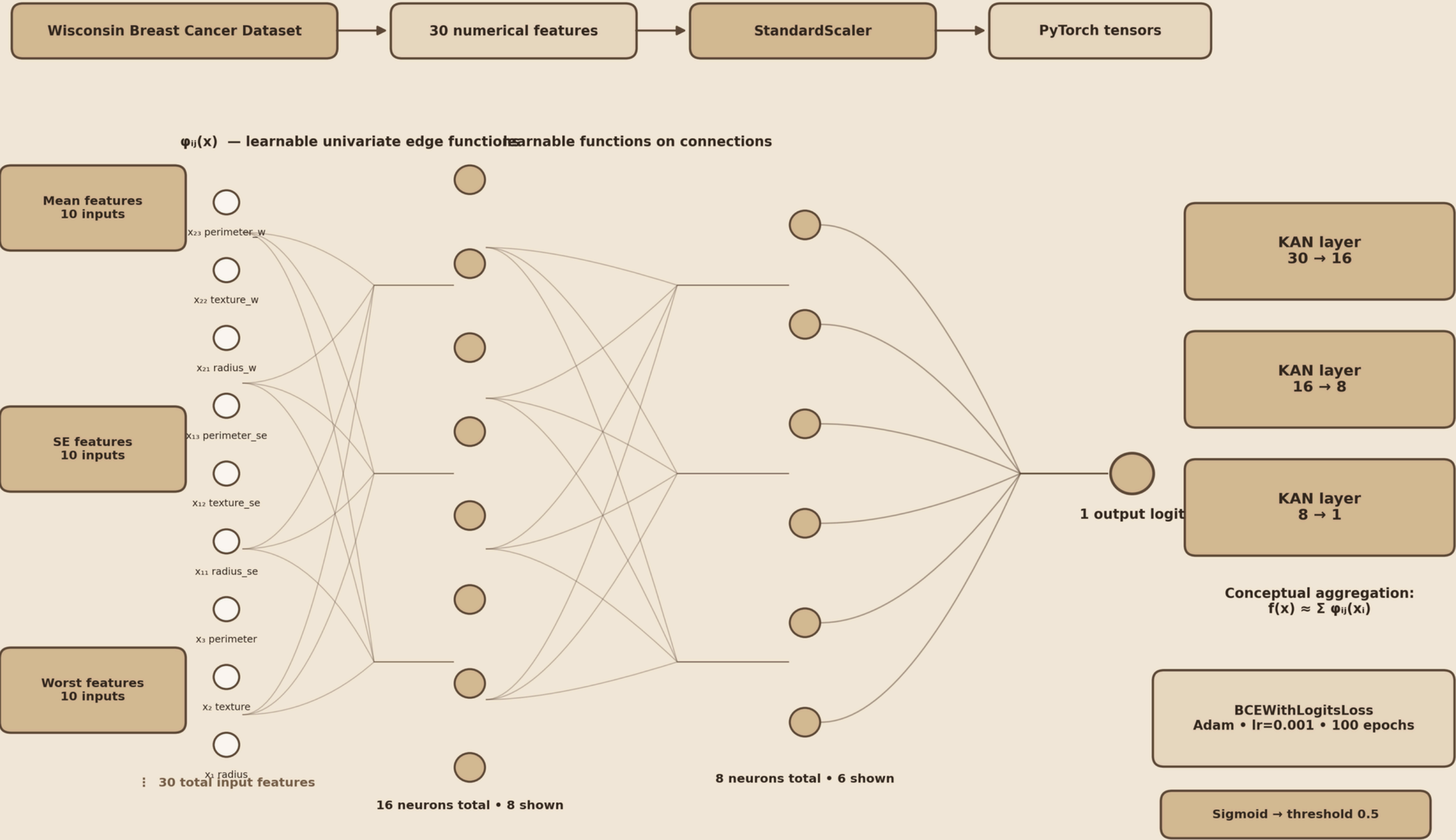


No hidden layers • one learned coefficient per feature + intercept

KAN

Detailed Kolmogorov-Arnold Network (KAN) Architecture

KAN(30,16,8,1) from your notebook — the internal edge-function view is shown conceptually.



	precision	recall	f1-score	support
Malignant	0.95	0.95	0.95	42
Benign	0.97	0.97	0.97	72
accuracy			0.96	114
macro avg	0.96	0.96	0.96	114
weighted avg	0.96	0.96	0.96	114

AutoEncoder

eg: let fixed $q.s = 100$

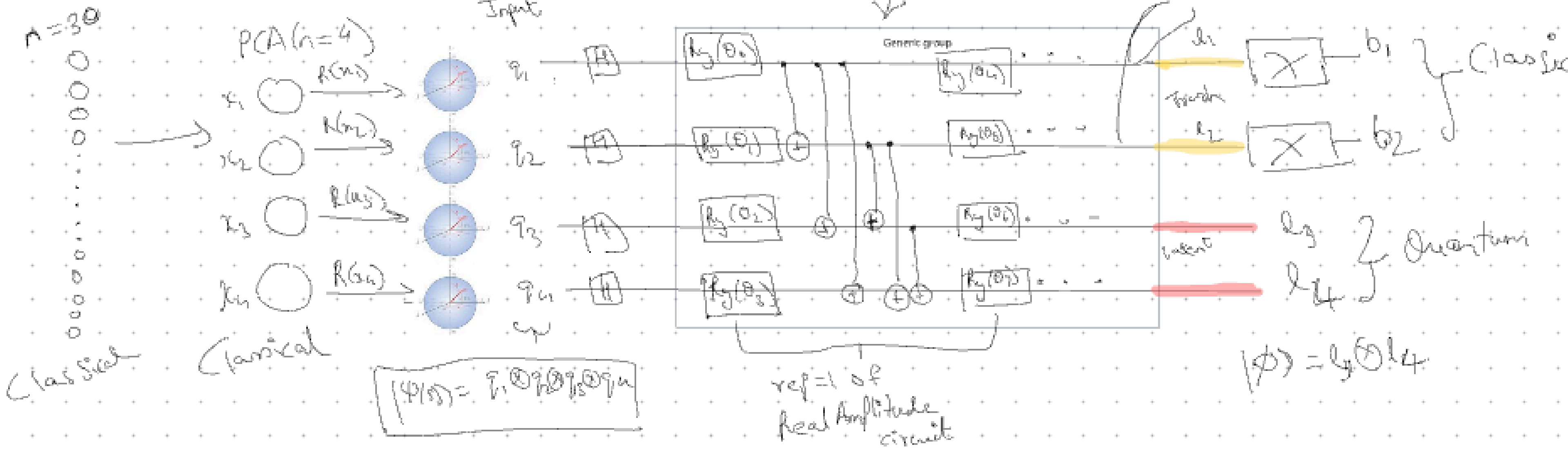
Threshold: 0.25
Accuracy: 0.9385964912288782

	precision	recall	f1-score	support
Malignant	0.97	0.86	0.91	42
Benign	0.92	0.99	0.95	72
accuracy			0.94	114
macro avg	0.95	0.92	0.93	114
weighted avg	0.94	0.94	0.94	114

$$\text{Fidelity} = |\langle 00|r \rangle|^2$$

$$\text{loss} = 1 - |\langle 00|r \rangle|^2$$

$|r\rangle = l_1 \otimes l_2 \leftarrow$ Trash quantum state



ROC-AUC: 0.9847883597883598

```
from sklearn.metrics import classification_report

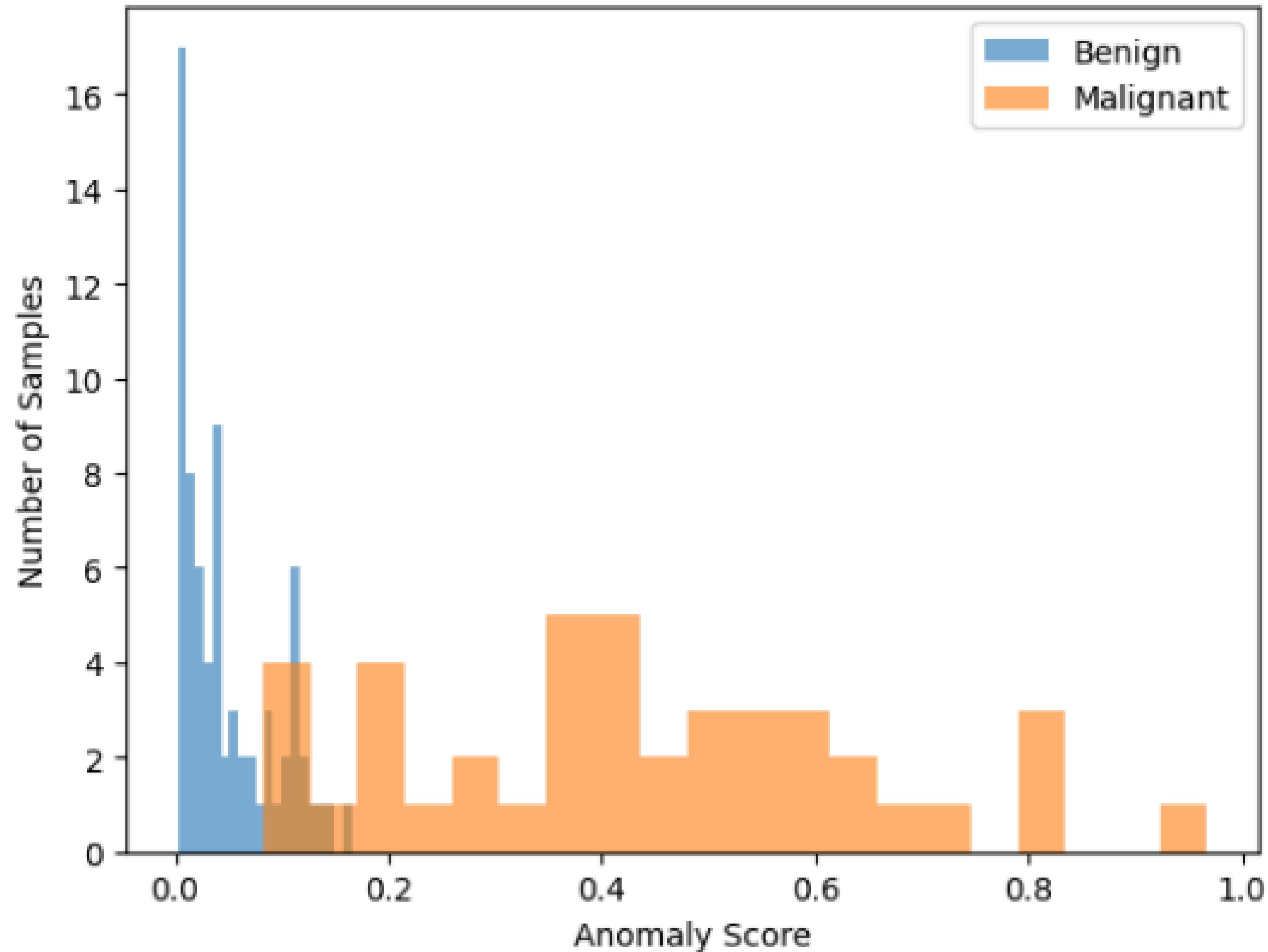
threshold = 0.14 # example threshold

# anomaly score > threshold => malignant
# anomaly score <= threshold => benign

y_pred = np.where(
    test_scores > threshold,
    0, # malignant
    1 # benign
)

print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Malignant", "Benign"]
    )
)
```

	precision	recall	f1-score	support
Malignant	0.95	0.90	0.93	42
Benign	0.95	0.97	0.96	72
accuracy			0.95	114
macro avg	0.95	0.94	0.94	114
weighted avg	0.95	0.95	0.95	114

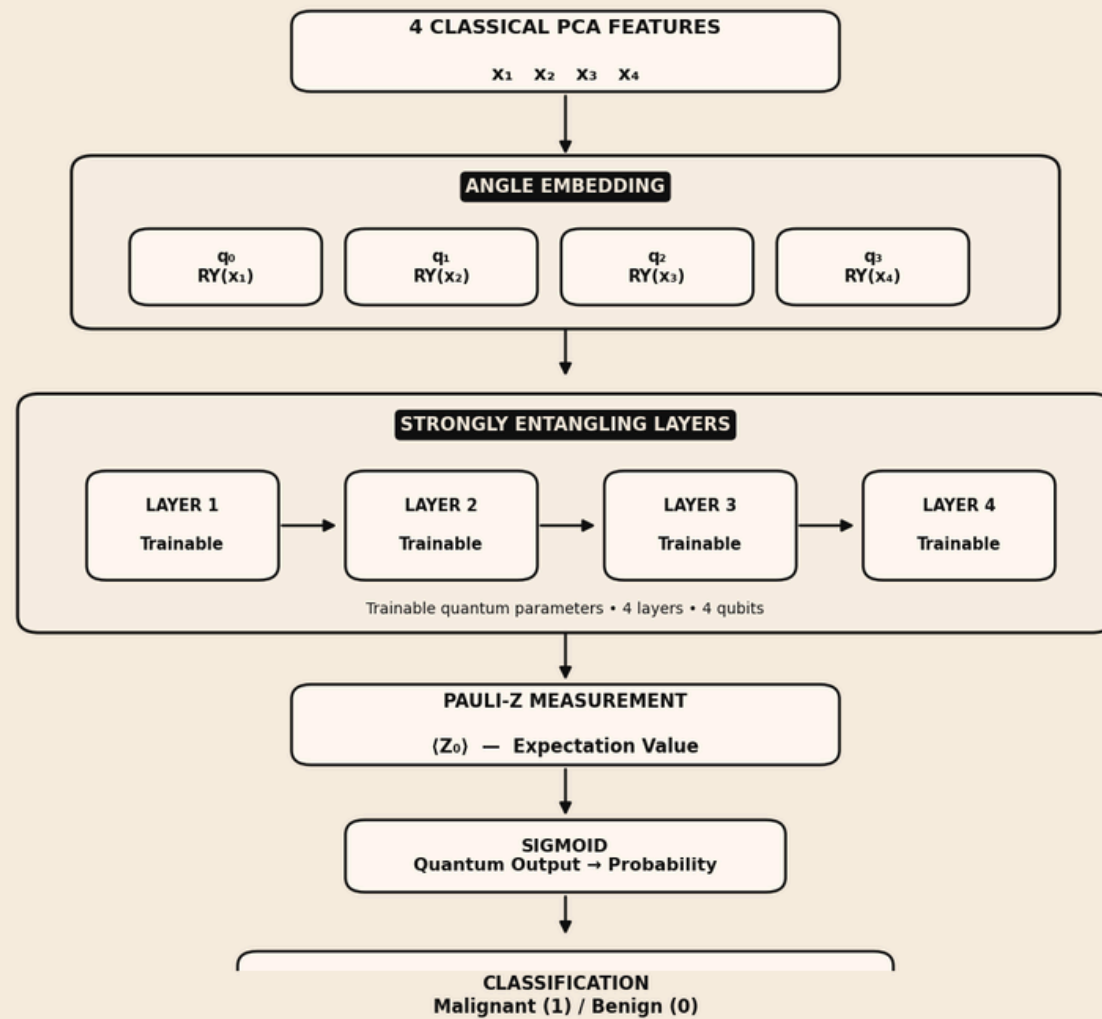


VQC

Accuracy : 0.9385964912280702
Precision: 0.9113924050632911
Recall : 1.0
F1 Score : 0.9536423841059603

INTERNAL ARCHITECTURE — QUANTUM LOGISTIC REGRESSION

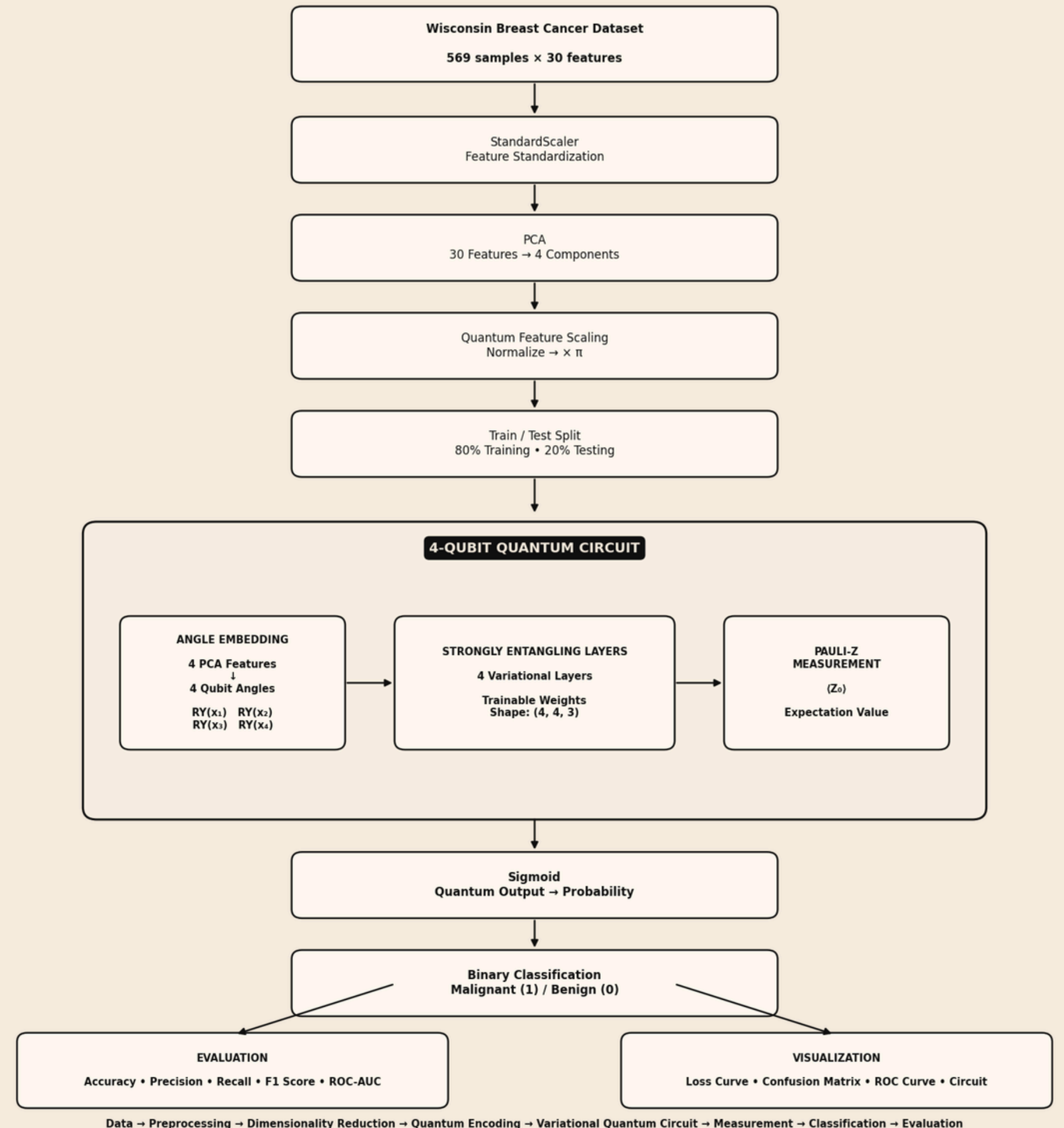
4-Qubit Variational Quantum Circuit



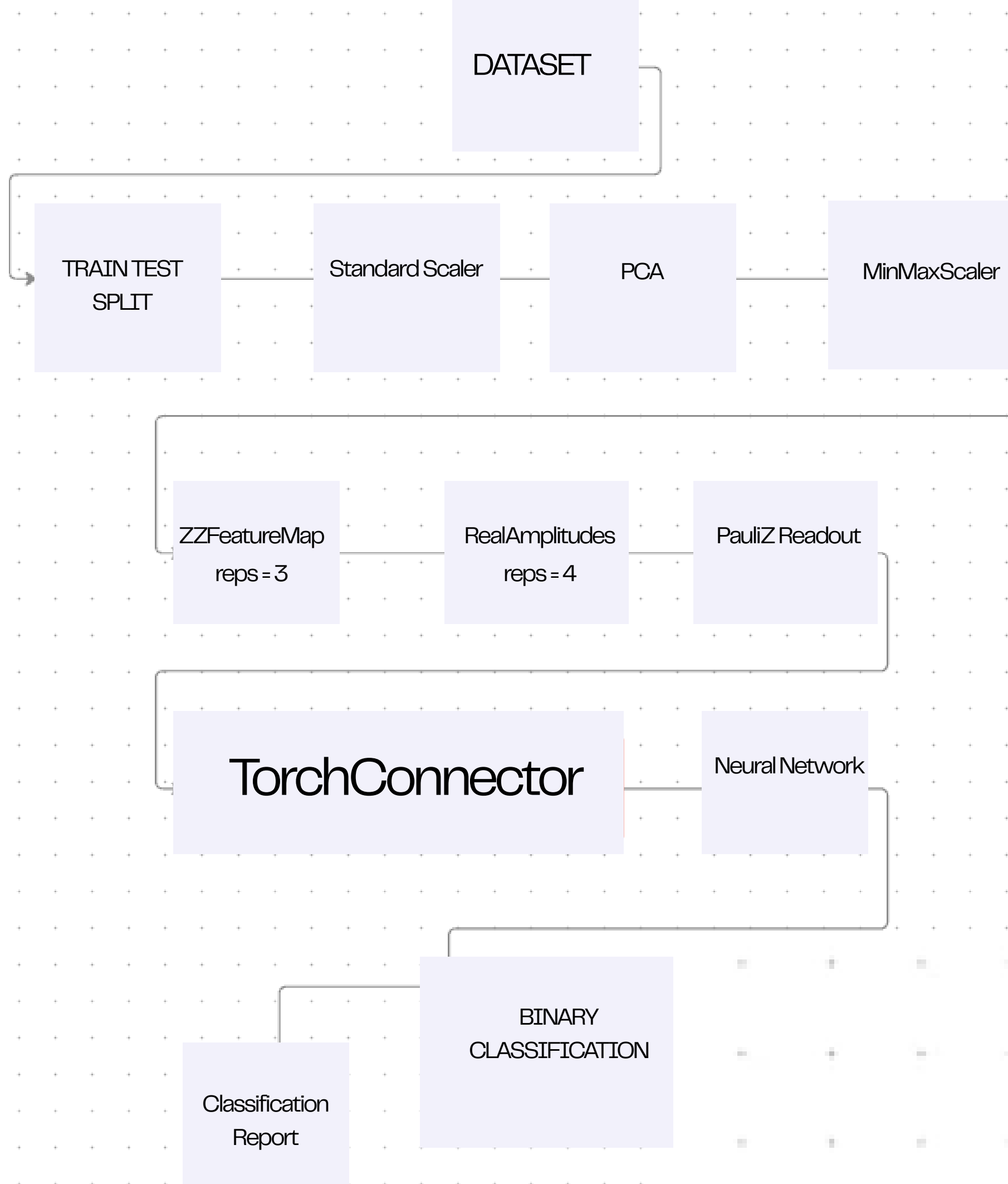
PCA Features → Angle Embedding → Trainable Quantum Layers → Pauli-Z → Sigmoid → Classification

QUANTUM LOGISTIC REGRESSION — MODEL ARCHITECTURE

Wisconsin Breast Cancer Dataset • PennyLane + PyTorch



Data → Preprocessing → Dimensionality Reduction → Quantum Encoding → Variational Quantum Circuit → Measurement → Classification → Evaluation

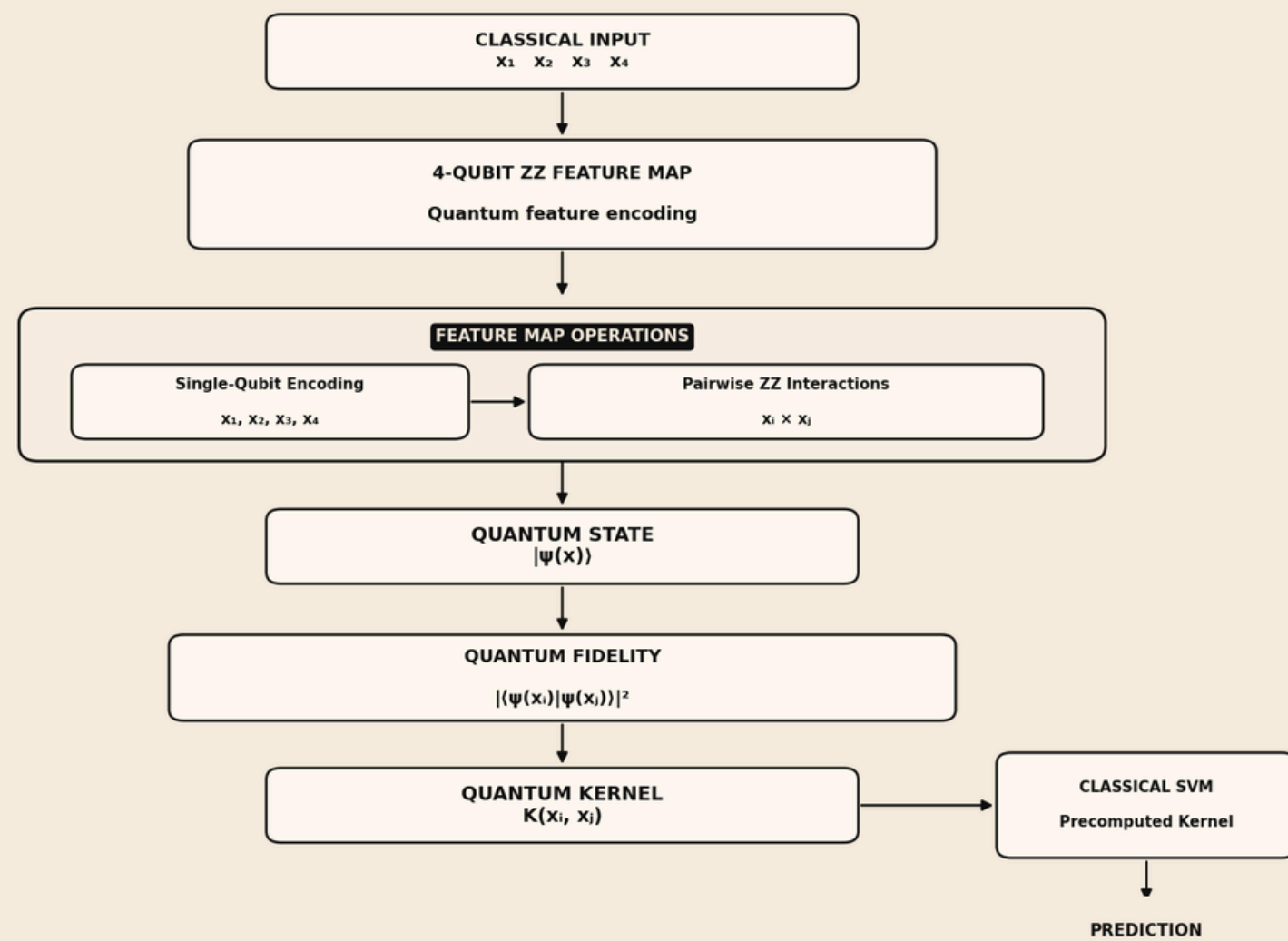


QSVC

	precision	recall	f1-score	support
malignant	0.95	0.83	0.89	42
benign	0.91	0.97	0.94	72
accuracy			0.92	114
macro avg	0.93	0.90	0.91	114
weighted avg	0.92	0.92	0.92	114

INTERNAL ARCHITECTURE — QUANTUM KERNEL / QSVM

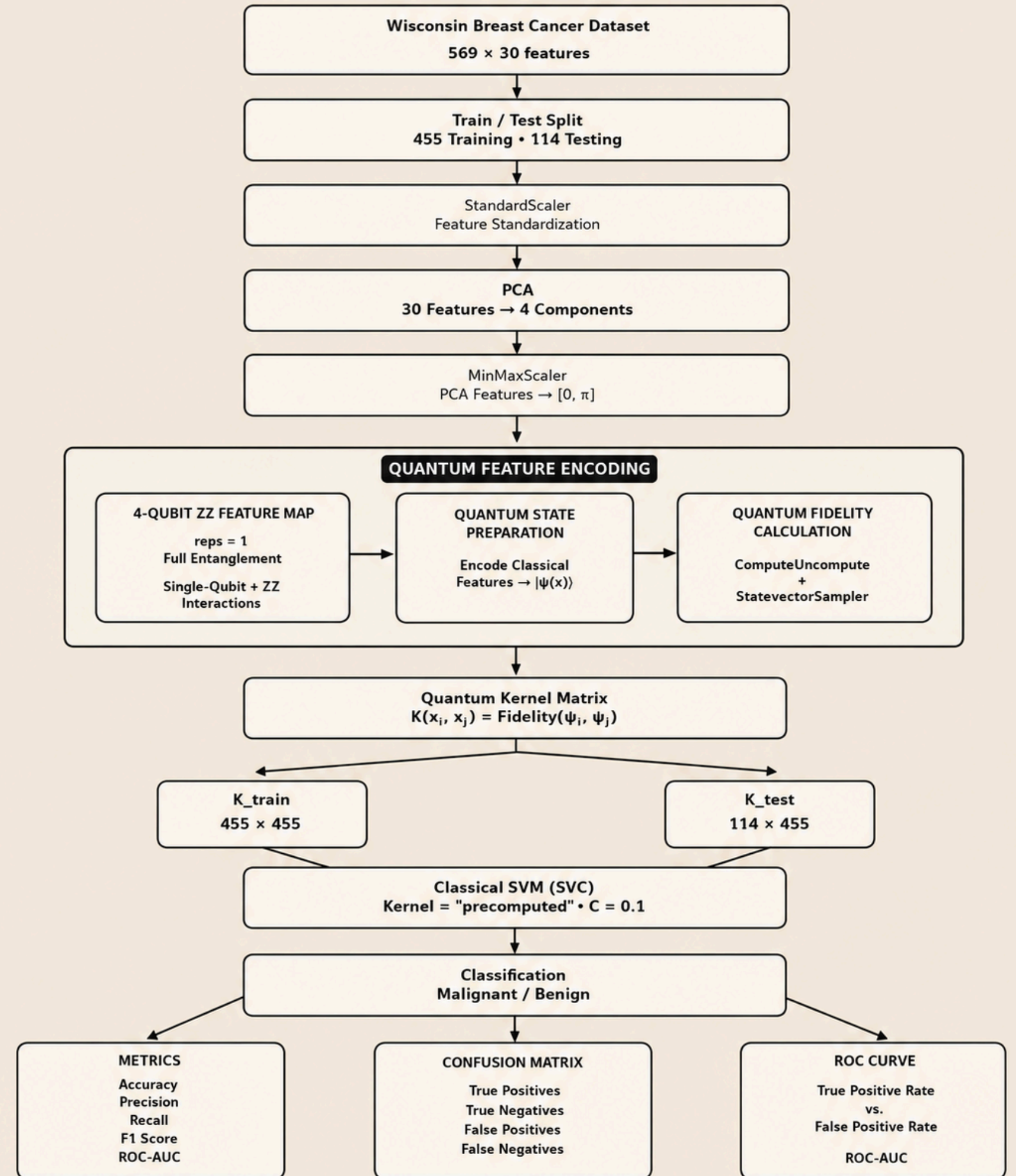
From classical features to quantum similarity and SVM prediction



Classical Features → Quantum Encoding → Quantum State → Fidelity → Quantum Kernel → Classical SVM → Prediction

QUANTUM SUPPORT VECTOR MACHINE — MODEL ARCHITECTURE

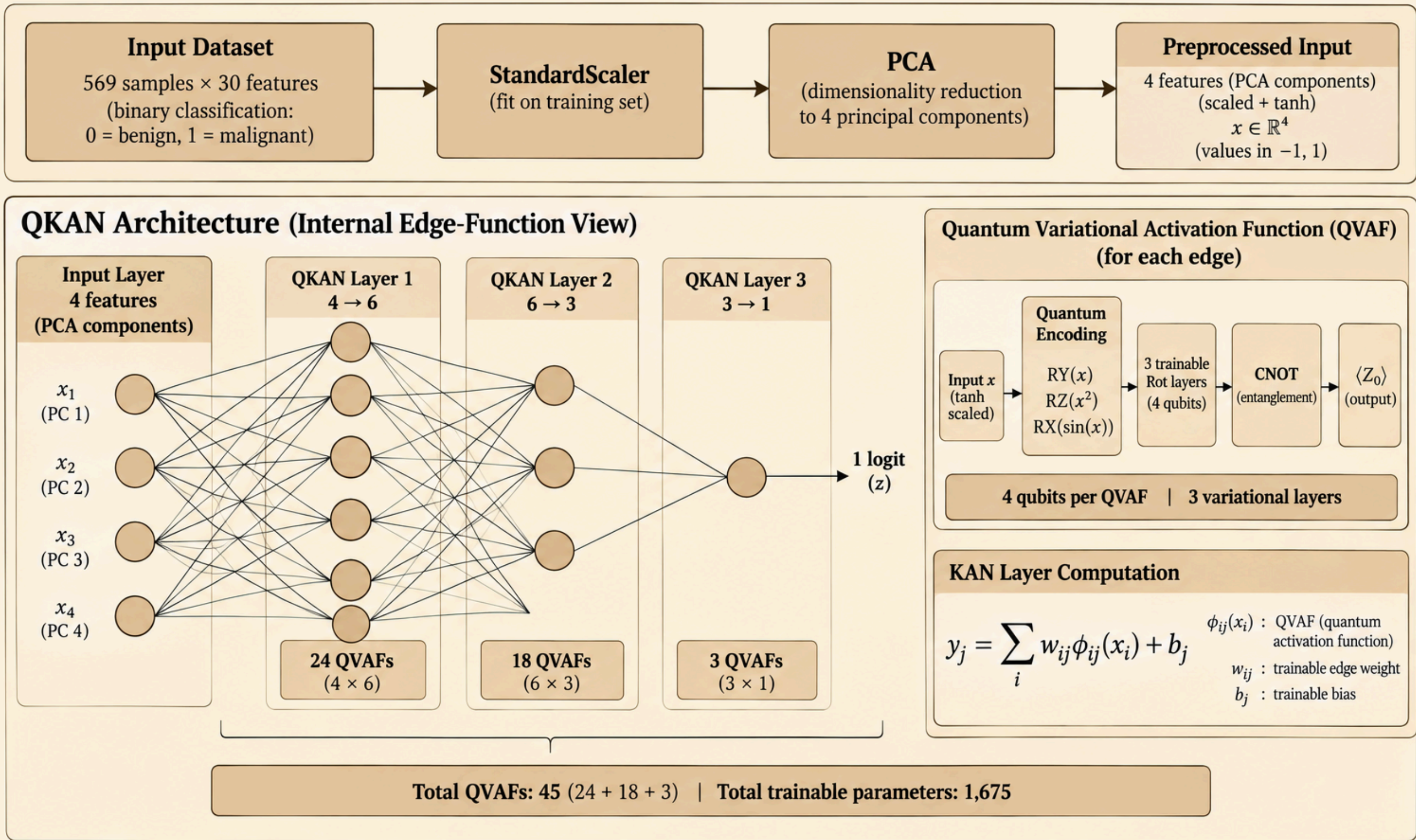
Wisconsin Breast Cancer Dataset • Qiskit Quantum Kernel



QKAN (QVAF)

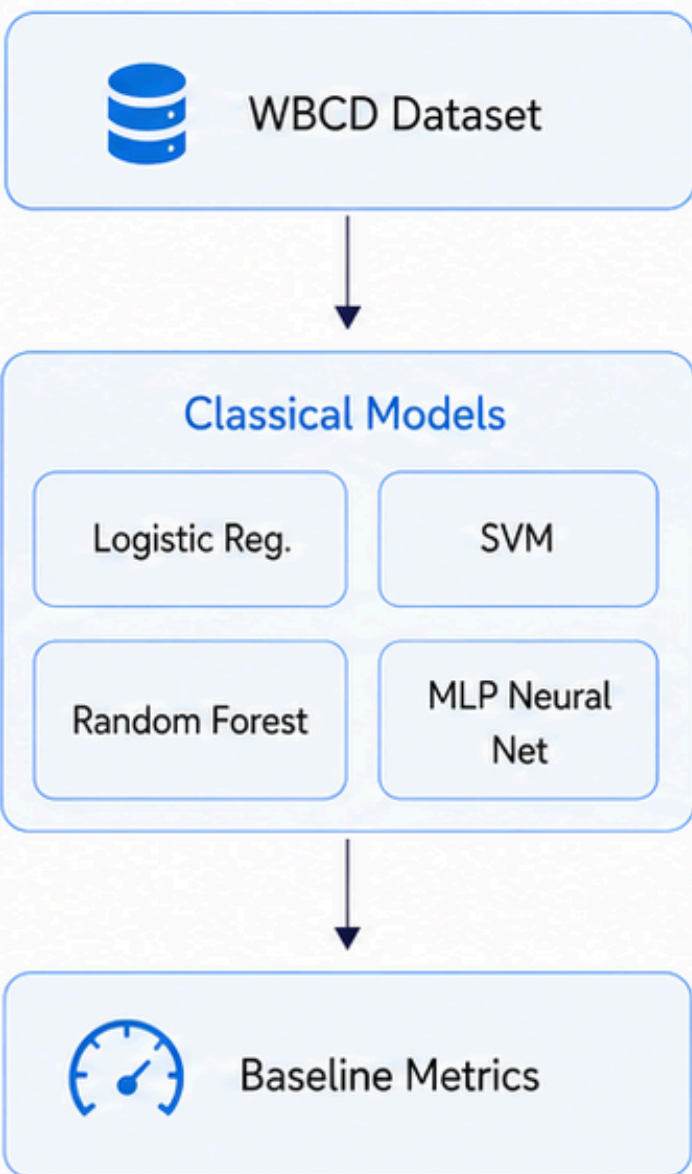
Classification Report:				
	precision	recall	f1-score	support
Benign	1.0000	0.9815	0.9907	54
Malignant	0.9697	1.0000	0.9846	32
accuracy			0.9884	86
macro avg	0.9848	0.9907	0.9876	86
weighted avg	0.9887	0.9884	0.9884	86

Quantum Kolmogorov-Arnold Network (QKAN) Architecture



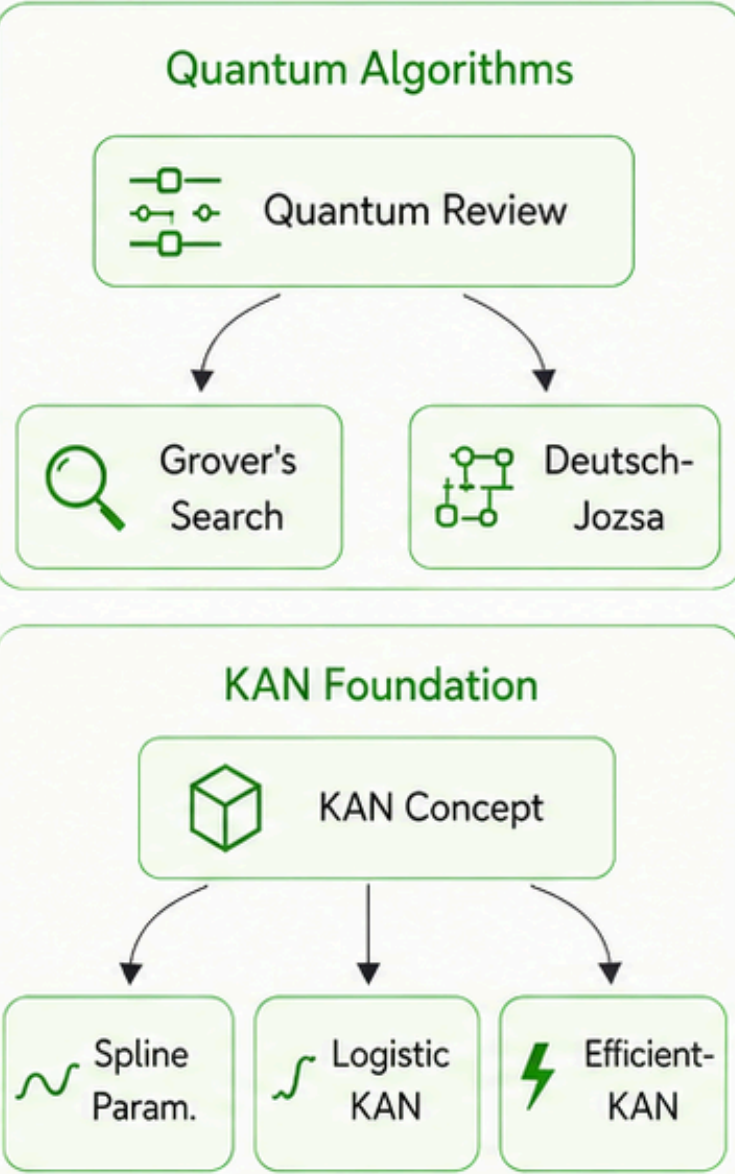
BIWEEKLY 1

Classical ML Pipeline



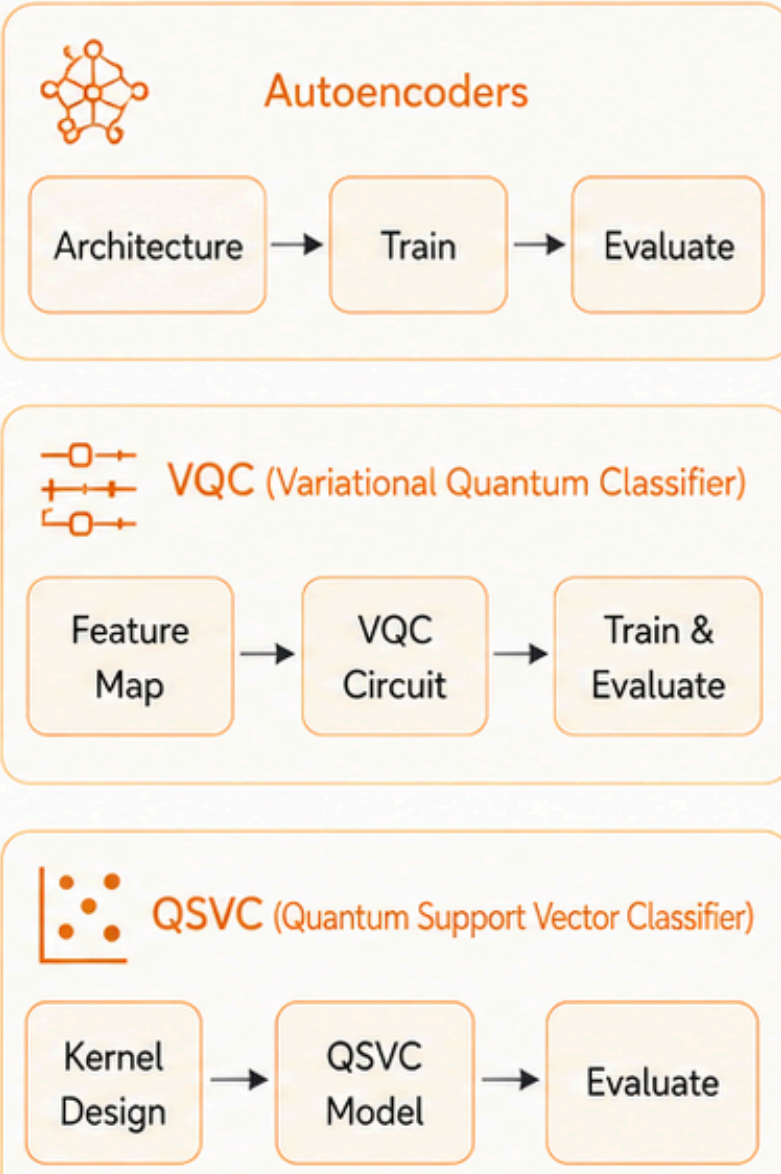
BIWEEKLY 2

KAN Theory & Learning Quantum



BIWEEKLY 3

Quantum Models



BIWEEKLY 4

Unified Comparison & Thesis

